Online Resource 1: Supplementary Information for "Participation Is Not Collaboration: Tracing Public Ownership After Cross-Product Agent Review"

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Abstract

This appendix gives the complete data contracts, joins, cohort rules, estimands, robustness checks, falsification tests, and experiment decisions for the accompanying article. It separates observed GitHub events from derived labels and keeps semantic and causal claims outside the evidence available in the public trace. All reported tables are generated from the validated analysis outputs rather than typed independently.

Keywords: coding agents, multi-agent software engineering, code review, supplementary information, reproducibility

Journal: Empirical Software Engineering **File:** Online Resource 1

1 Purpose and interpretation boundary

The main article keeps only the numbers needed to understand the story. This appendix carries the complete measurement and sensitivity record. The study observes public GitHub accounts and events. A mapped product label does not reveal its foundation model, private messages, or internal run boundary. A GitHub `User` account does not prove unaided manual work. A later event does not prove that a review point was understood or fixed. All outcome models are observational associations.

We use four evidence levels. *Product presence* means that two mapped products appear on one pull request (PR). An *addressed edge* requires a reply whose parent identifier is the exact cross-product trigger. A *next owner* is the first public actor class after a declared rapid-burst window. A *later state* is measured only after a fixed landmark. These levels must not be collapsed into a single "multi-agent" indicator.

This boundary also separates the study from prior work on cross-product review prevalence, broad reviewer-type sequences, and developer response (Selvanayagam and Ghaleb 2026; Zhong et al. 2026; Cynthia et al. 2026). Those studies motivate the narrower event-level checks reported here.

2 Dataset release, grain, and joins

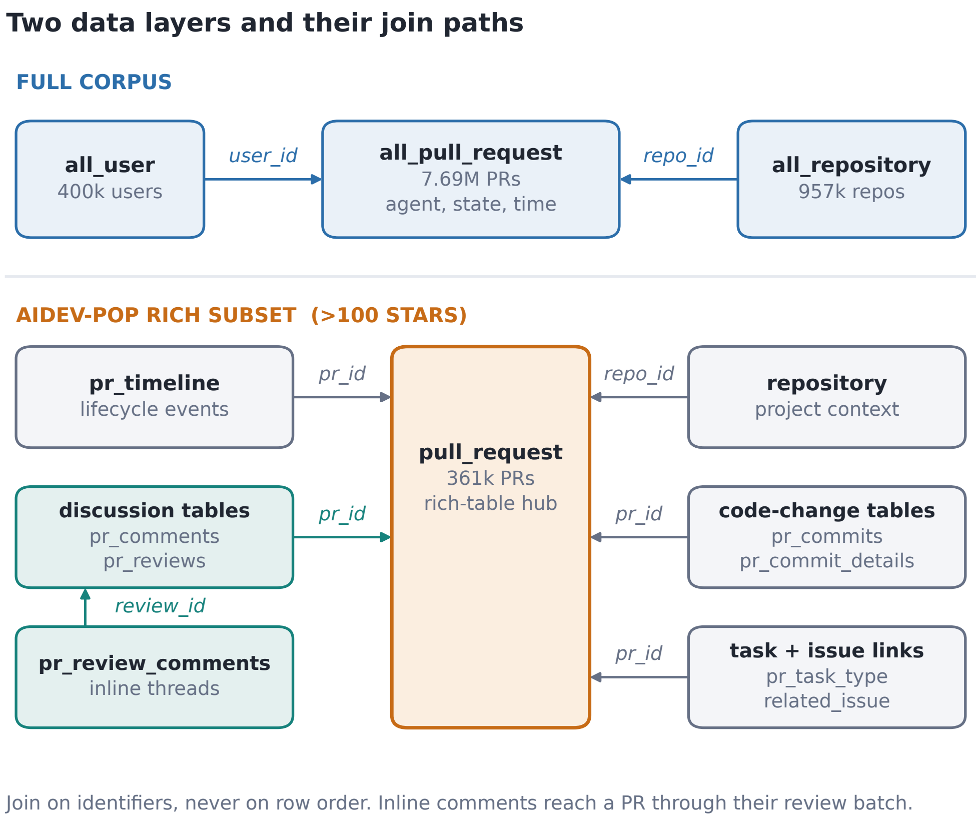

Fig. 1 AIDev release layers and the identifier paths used in this study. The full corpus supports broad PR, repository, and account checks. The rich layer supports event-level analysis. Inline review comments reach a PR through their submitted-review identifier; row order is never a join key.

The analysis pins AIDev-7.6M at revision 37bbe1533e26cc1e1374917dba1186d1c8a4dc81. The full release is used for corpus checks. Interaction analyses use AIDev-pop, the rich layer for repositories with more than 100 stars. Table 1 reports the release profile and Table 2 reports event-table coverage.

The PR table is the hub. Submitted reviews join directly by PR identifier. Inline review comments do not contain a PR identifier, so they first join to a submitted review by `pull_request_review_id`; the review then joins to the PR. All inline-review identifiers used in this join resolve. PR comments and timeline events join directly to the PR. Commit-detail rows have one row per file and commit; any PR-level commit count must deduplicate commit SHA values before aggregation.

Figure 1 shows the two release layers and their main join paths. Figure 2 shows why each analysis declares its own denominator: rich tables cover different subsets of the same PR backbone.

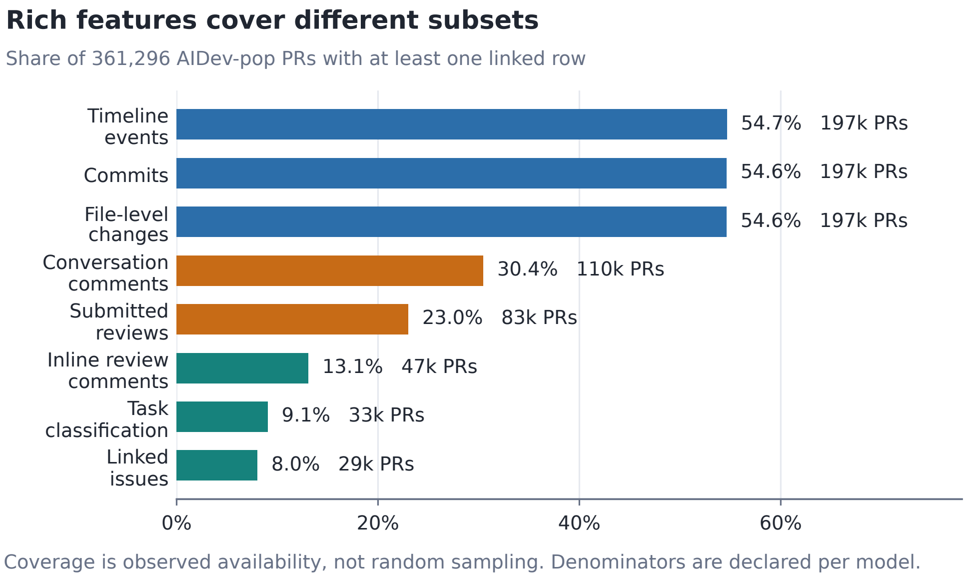

Fig. 2 Availability of rich feature groups across the 361,296-PR backbone. Coverage is not uniform and missing linked rows are not assumed to be negative events. Each model therefore uses a declared feature-specific denominator.

3 Complete quantitative record

The following tables collect every reported denominator, estimate, robustness check, and experiment decision. Later sections explain how each table was built and what it can and cannot support.

Table 1: Dataset tables, units, and analytical roles in the pinned release.

Table	Scope	Unit	Rows	Fields	Role
all_pull_request	Full corpus	One pull request	7,685,281	14	PR identity, agent, author, state, timestamps, and repository
all_repository	Full corpus	One repository	957,209	8	Repository name, language, license, fork status, stars, and forks
all_user	Full corpus	One GitHub user	399,962	5	Contributor account age and follower/following counts
pull_request	Rich layer	One pull request	361,296	14	PR backbone for the richer event and content tables
repository	Rich layer	One repository	25,402	8	Repository context for the richer subset
pr_timeline	Rich layer	One PR timeline event	3,018,358	8	Review, assignment, label, closure, and other lifecycle events
pr_comments	Rich layer	One issue-style PR comment	373,549	7	Conversation text, actor, actor type, and time
pr_reviews	Rich layer	One submitted PR review	281,170	8	Review decision, reviewer type, text, and submission time
pr_review_comment s	Rich layer	One inline review comment	289,780	15	Inline discussion tied to a file and diff position
pr_commits	Rich layer	One commit attached to a PR	718,779	5	Commit author, committer, SHA, and message
pr_commit_details	Rich layer	One changed file within a PR commit	6,112,623	14	Change size, filename, file status, and patch text
pr_task_type	Rich layer	One classified pull request	32,702	17	Task type, model reason, and classification confidence
issue	Rich layer	One linked issue	29,111	9	Issue title, body, author, state, and timestamps

Dataset tables, units, and analytical roles in the pinned release. (continued)

Table	Scope	Unit	Rows	Fields	Role
related_issue	Rich layer	One PR-issue link	38,006	3	Bridge between PRs and issues, with link source

Note: The rich layer contains repositories with more than 100 stars. Row counts come from Parquet metadata, so large content tables are not expanded in memory.

Table 2 Coverage of rich event and feature tables across 361,296 AIDev-pop pull requests.

Feature group	Rows	Matched PRs	PR coverage (%)	Orphan PR IDs
Timeline events	3,018,358	197,471	54.66	0
File-level changes	6,112,623	197,330	54.62	0
Commits	718,779	197,335	54.62	0
Conversation comments	373,549	110,011	30.45	0
Submitted reviews	281,170	83,081	23.00	0
Inline review comments	289,780	47,251	13.08	0
Task classification	32,702	32,702	9.05	0
Linked issues	38,006	28,987	8.02	0

Note: Coverage means that a pull request has at least one linked row. Inline comments resolve through their submitted-review identifier before joining to the pull request.

Table 3 Exact public-account allowlist used for reviewer-product attribution.

Mapped product	Exact GitHub login(s), matched case-insensitively
`Claude_Code`	`claude[bot]`
`Copilot`	`copilot`, `copilot-pull-request-reviewer[bot]`, `copilot-swe-agent[bot]`
`Cursor`	`cursor[bot]`
`Devin`	`devin-ai-integration[bot]`
`Google_Jules`	`google-labs-jules[bot]`
`OpenAI_Codex`	`chatgpt-codex-connector[bot]`

Note: Only exact aliases are mapped. Similar names and unknown bots stay outside the six-product registry.

Table 4: Glossary of load-bearing terms, read from the analysis code.

Term	Definition in the code	Where it is defined
Decisive review	A submitted review whose state is APPROVED or CHANGES_REQUESTED. Commented and pending reviews are not decisive.	Landmark cohort builder
Branch movement	A `head_ref_force_pushed` timeline event on the PR. It is visible movement of the branch, not a verified repair.	Timeline event filter
Untyped activity	A post-trigger event row that is not a user account, a mapped product, or another bot. It is the residual class and is always reported together with branch movement.	First-state classifier
State: user account	The event's GitHub user type is User. User evidence takes precedence over product mapping.	First-state classifier
State: mapped product	The event is not a User event and its account is on the six-product allowlist.	First-state classifier
State: other bot	The event's user type is Bot and the account is not on the allowlist.	First-state classifier
State: branch movement or untyped activity	Every remaining event row, which in this release is branch movement and untyped rows.	First-state classifier
State: no later action within seven days	No post-burst event occurs inside the seven-day response window.	First-state classifier
Role: the PR author's own user account	A User account whose login equals the PR author's login.	Response actor roles
Role: the PR author's product	A mapped product equal to the product attributed to the PR author.	Response actor roles
Role: the triggering product	A mapped product equal to the product that wrote the trigger.	Response actor roles
Role: a different mapped product	Any other account on the six-product allowlist.	Response actor roles
Role: an unmapped bot	A Bot account that is not on the allowlist.	Response actor roles
Role: another user account	Any other User account.	Response actor roles
Role: unknown actor	An event with no usable user type. The class is kept so that the seven roles stay exhaustive.	Response actor roles
Automation episode	A run of post-burst events on one PR in which each event follows the previous one by no more than the gap parameter; a larger gap starts a new episode. The primary gap is five minutes, and the first five minutes after the trigger are excluded before episodes are built.	Episode sessionization

Glossary of load-bearing terms, read from the analysis code. (continued)

Term	Definition in the code	Where it is defined
Episode state	The highest-priority state inside the episode, using the same priority order as the tie rule. At the five-minute gap, 2,529 of 14,842 episodes mix states.	Episode sessionization

Note: Each definition is the rule the code applies, not an interpretation of it. The five states are mutually exclusive and cover every PR at every burst threshold. The seven roles are evaluated in the listed order, so the first matching branch wins. None of these labels reports intent, private activity, or whether a review point was resolved.

Table 5 Participation-to-addressed-edge denominator funnel.

Stage	PRs	Denominator	Share (%)
Complete seven-day trigger cohort	8,608	Full cohort	100.0
Any later visible action	5,553	Full cohort	64.5
Inline trigger; exact reply is observable	4,824	Full cohort	56.0
Exact parent reply	730	Full cohort	8.5
Exact parent reply	730	Inline-eligible	15.1
Mapped different-product exact reply	74	Full cohort	0.9
Mapped different-product exact reply	74	Inline-eligible	1.5

Note: An exact parent reply is defined only for an inline trigger. The repeated rows make the full-cohort and eligible denominators explicit.

Table 6: Named estimation specifications behind the reported outcome models.

Specification	Outcome	Exposure term	Estimator and errors	PRs / repos	Formula
Exact edge, pre-trigger adjusted (primary)	`merged_from_48h_to_30d`	`exact_parent_reply_by_48h`	Linear probability model by OLS; cluster-robust by repository	1,067 / 469	`merged_from_48h_to_30d ~ exact_parent_reply_by_48h + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes`
Exact edge, route decomposition (secondary)	`merged_from_48h_to_30d`	`exact_parent_reply_by_48h`	Linear probability model by OLS; cluster-robust by repository	1,067 / 469	`merged_from_48h_to_30d ~ exact_parent_reply_by_48h + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes + C(ownership_route_48h, Treatment('no_observed_action'))`
Specificity: edge vs non-exact discussion	`merged_from_48h_to_30d`	`exact_edge_by_48h`	Linear probability model by OLS; cluster-robust by repository	615 / 300	`merged_from_48h_to_30d ~ exact_edge_by_48h + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes`
Specificity: edge vs any other visible activity	`merged_from_48h_to_30d`	`exact_edge_by_48h`	Linear probability model by OLS; cluster-robust by repository	641 / 309	`merged_from_48h_to_30d ~ exact_edge_by_48h + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes`

Named estimation specifications behind the reported outcome models. (continued)

Specification	Outcome	Exposure term	Estimator and errors	PRs / repos	Formula
Specificity: user edge vs non-exact user discussion	merged_from_48h_to_30d	exact_user_edge_by_48h	Linear probability model by OLS; cluster-robust by repository	369 / 215	merged_from_48h_to_30d ~ exact_user_edge_by_48h + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes
Four-state response gradient	merged_from_48h_to_30d	specificity_group_48h	Linear probability model by OLS; cluster-robust by repository	1,067 / 469	merged_from_48h_to_30d ~ C(specificity_group_48h, Treatment('no_visible_activity')) + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes
Ownership route, base controls	merged_from_48h_to_30d	ownership_route_48h	Linear probability model by OLS; cluster-robust by repository	1,733 / 641	merged_from_48h_to_30d ~ C(ownership_route_48h, Treatment('automation_no_human')) + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_source) + C(trigger_month)
Ownership route, pre-trigger adjusted	merged_from_48h_to_30d	ownership_route_48h	Linear probability model by OLS; cluster-robust by repository	1,733 / 641	merged_from_48h_to_30d ~ C(ownership_route_48h, Treatment('automation_no_human')) + C(author_agent) + C(trigger_reviewer_agent) + C(trigger_source) + C(trigger_month) + log1p_trigger_age_hours + log1p_pre_events + pre_user_events + pre_bot_events + pre_decisive_reviews + pre_force_pushes

Named estimation specifications behind the reported outcome models. (continued)

Specification	Outcome	Exposure term	Estimator and errors	PRs / repos	Formula
Prior history: first user bridge	`prior_different_pr_reviewer`	`other_user`	Linear probability model by OLS; cluster-robust by repository, finite-sample corrected	3,603 / 834	`prior_different_pr_reviewer ~ other_user`
Prior history: all 48-hour responders	`prior_different_pr_reviewer`	`is_first`	Linear probability model by OLS; cluster-robust by repository, finite-sample corrected	4,814 / 834	`prior_different_pr_reviewer ~ is_first + is_author`

Note: Every row is a linear probability model fitted by ordinary least squares, with standard errors clustered on the repository. Formulas are stored with the analysis products themselves, except the two route rows and the two prior-history rows, whose formulas are read back from the analysis code. The reply-window variants at 1, 6, and 24 hours refit the same formula with the exposure term for that window. A specification describes the fit; it does not claim that the fit identifies an effect.

Table 7: First public state after each rapid-burst threshold.

Burst (min)	First state	PRs	Share (%)	Cluster 95% interval	Median min
0	User account	2,381	27.7	[23.6, 32.5]	26.8
0	Mapped product	1,305	15.2	[12.7, 17.8]	4.5
0	Other bot	1,359	15.8	[12.4, 19.6]	2.5
0	Branch movement or untyped activity	508	5.9	[4.4, 7.5]	30.9
0	No later action within seven days	3,055	35.5	[30.7, 40.5]	–
1	User account	2,428	28.2	[24.2, 32.7]	31.1
1	Mapped product	1,198	13.9	[11.9, 16.1]	5.7
1	Other bot	1,161	13.5	[10.6, 16.7]	3.4
1	Branch movement or untyped activity	552	6.4	[4.9, 8.1]	33.6
1	No later action within seven days	3,269	38.0	[33.3, 42.8]	–
5	User account	2,526	29.3	[25.3, 33.8]	41.5
5	Mapped product	924	10.7	[9.1, 12.4]	14.6
5	Other bot	761	8.8	[6.8, 11.3]	6.9
5	Branch movement or untyped activity	560	6.5	[4.9, 8.3]	45.1
5	No later action within seven days	3,837	44.6	[40.1, 49.1]	–
10	User account	2,413	28.0	[24.1, 32.2]	61.9
10	Mapped product	827	9.6	[8.1, 11.2]	17.9
10	Other bot	609	7.1	[5.4, 9.3]	11.0
10	Branch movement or untyped activity	554	6.4	[4.8, 8.1]	56.8
10	No later action within seven days	4,205	48.8	[44.6, 53.2]	–
30	User account	2,165	25.2	[21.9, 28.8]	134.9
30	Mapped product	653	7.6	[6.4, 8.9]	41.4
30	Other bot	377	4.4	[3.5, 5.4]	45.7
30	Branch movement or untyped activity	483	5.6	[4.3, 7.1]	117.8
30	No later action within seven days	4,930	57.3	[53.2, 61.1]	–

Note: The denominator is 8,608 PRs at every threshold. The window is a sensitivity parameter, not an inferred private session boundary.

Table 8 Tied first timestamps under the fixed state priority order.

Burst threshold	PRs with a post-burst action	Tied first timestamp	Mixed-state ties	Largest tie group
0 minutes	5,553	366	1	10
1 minutes	5,339	372	0	10
5 minutes	4,771	306	0	10
10 minutes	4,403	241	0	10
30 minutes	3,678	169	0	10

Note: The priority order is user account, then mapped product, then other bot, then branch movement or untyped activity. Ties are common because many events share a whole-second timestamp, but a tie changes the assigned state only when the tied events fall in different states. Exactly one mixed-state tie occurs, at the zero-minute threshold, and none occurs at any positive threshold. The order favours the user-account state, which is the state the RQ1 ordering claim rests on, so the diagnostic is reported rather than assumed harmless.

Table 9: Leave-one-out ordering robustness for the first post-burst state.

Burst (min)	Exclusion unit	Quantity	Full (%)	Leave-one-out range	Gate or largest shift
0	Repository	User minus mapped product	–	10.5 to 13.2 pp	both gates hold in all 1,416 exclusions
1	Repository	User minus mapped product	–	12.3 to 15.1 pp	both gates hold in all 1,416 exclusions
5	Repository	User minus mapped product	–	16.9 to 19.1 pp	both gates hold in all 1,416 exclusions
10	Repository	User minus mapped product	–	16.8 to 18.8 pp	both gates hold in all 1,416 exclusions
30	Repository	User minus mapped product	–	16.3 to 17.9 pp	both gates hold in all 1,416 exclusions
0	Ordered product pair	User minus mapped product	–	10.2 to 14.2 pp	both gates hold in all 25 exclusions
1	Ordered product pair	User minus mapped product	–	11.6 to 16.0 pp	both gates hold in all 25 exclusions
5	Ordered product pair	User minus mapped product	–	16.5 to 19.9 pp	both gates hold in all 25 exclusions
10	Ordered product pair	User minus mapped product	–	17.0 to 19.2 pp	both gates hold in all 25 exclusions
30	Ordered product pair	User minus mapped product	–	16.5 to 18.4 pp	both gates hold in all 25 exclusions
5	Repository	User account	29.3	27.9 to 29.9	largest shift 1.43 pp
5	Repository	Mapped product	10.7	10.6 to 11.0	largest shift 0.31 pp
5	Repository	Other bot	8.8	8.0 to 9.0	largest shift 0.87 pp
5	Repository	Branch movement or untyped activity	6.5	6.0 to 6.7	largest shift 0.47 pp
5	Repository	No later action within seven days	44.6	43.5 to 45.4	largest shift 1.10 pp
5	Ordered product pair	User account	29.3	28.0 to 30.4	largest shift 1.34 pp
5	Ordered product pair	Mapped product	10.7	10.1 to 13.9	largest shift 3.16 pp
5	Ordered product pair	Other bot	8.8	7.7 to 9.2	largest shift 1.18 pp
5	Ordered product pair	Branch movement or untyped activity	6.5	5.4 to 6.9	largest shift 1.07 pp

Leave-one-out ordering robustness for the first post-burst state. (continued)

Burst (min)	Exclusion unit	Quantity	Full (%)	Leave-one-out range	Gate or largest shift
5	Ordered product pair	No later action within seven days	44.6	41.2 to 45.2	largest shift 3.42 pp

Note: The first block drops one repository, or one ordered author–reviewer product pair, at a time and refits the ordering. Two gates are checked in every exclusion: the user-account share stays above the mapped-product share, and no later action within seven days stays the largest state. The second block reports the share range for each state at the primary five-minute threshold. These checks show that the ordering is not carried by one repository or one product pair. They do not remove selection into the cohort.

Table 10 Deep transition after a mapped-product first state and a second five-minute washout.

Quantity	PRs	Estimate	Repository-cluster 95% interval
Next state: User account	217	23.5	[19.7, 27.4]
Next state: Same mapped product	284	30.7	[26.6, 35.0]
Next state: Different mapped product	43	4.7	[3.2, 6.2]
Next state: Other bot	39	4.2	[2.6, 6.1]
Next state: Branch movement or untyped activity	36	3.9	[2.4, 5.4]
Next state: No later state	305	33.0	[29.0, 37.2]
Same-product continuation among later mapped events: observed	495	84.4	–
Same-product continuation: random-order expectation	495	83.6	–
Observed minus random-order expectation (percentage points)	495	+0.9	[-1.1, +3.1]

Note: The next-state shares use 924 mapped-first PRs. The placebo uses 495 PRs with a later mapped-product event and preserves each PR's future product composition.

Table 11 Falsification tests for escalation and persistent ownership stories.

Test	At risk	Observed (%)	Null or difference (pp)	95% interval
Automation episode 1: next is a user	1,742	25.8	25.1	[23.7, 26.7]
Automation episode 2: next is a user	666	19.1	17.0	[14.3, 19.5]
Automation episode 3: next is a user	341	15.8	12.6	[9.3, 16.1]
Automation episode 4: next is a user	205	9.8	10.2	[6.4, 14.0]
Visible next action: Exact owner repeats	3,237 start PRs	user 31.0; product 44.5	-13.5	[-20.4, -6.8]
Visible next action: Same ownership layer continues	3,237 start PRs	user 54.1; product 53.3	+0.7	[-6.7, +8.3]
Visible next action: Cross-layer handoff (user–product bounce)	3,237 start PRs	user 31.9; product 33.0	-1.1	[-7.6, +5.7]

Note: Automation rows compare the observed user-next share with a within-PR order permutation. Ownership rows condition on a visible next action; their fourth column is user-first minus mapped-first. Same-layer continuation and cross-layer bounce show no clear group difference, while exact owner repetition is higher after a mapped-product first state.

Table 12 Exact-author matched cross-product versus same-product trigger comparison.

Outcome	Pairs	Cross (%)	Same (%)	Difference (pp)	Repository-cluster 95% interval
Any visible follow-up	546	68.1	81.5	-13.4	[-19.9, -4.0]
Later PR comment	546	38.5	53.8	-15.4	[-33.2, +2.6]
New review round	546	52.6	51.8	+0.7	[-13.5, +13.3]
Exact trigger reply	546	7.5	8.4	-0.9	[-4.0, +1.8]
Visible force-push	546	11.0	9.2	+1.8	[-1.6, +4.8]
Merge within seven days	546	73.4	70.7	+2.7	[-2.9, +10.2]
Any visible follow-up; seven-day pairing caliper	423	67.6	82.5	-14.9	[-21.6, -4.2]

Note: Pairs also match repository, author product, trigger source, and month, then use nearest trigger time without replacement. Difference is cross-product minus same-product.

Table 13 Strict prior same-repository review history across user-account populations.

Population	Account–PR rows	Repos	With prior history (%)	Median prior PRs if experienced
First user bridge: PR author	1,372	438	67.1	7
First user bridge: another account	2,231	571	73.8	11
All first user bridges	3,603	834	71.2	9
First later decisive reviewer	2,020	488	78.2	11
All distinct 48-hour user responders	4,814	834	70.2	9
Cross-feedback PR author accounts	8,608	1,416	40.4	4

Note: Prior history requires the same account, the same repository, a different PR, and a submitted review strictly before the cross-product trigger. It is observable public history, not verified memory retrieval.

Table 14 Exact-parent addressed edge and later merge from the 48-hour landmark through day 30.

Reply window	Specification	Edge PRs	Raw merge: edge / no edge (%)	Adjusted difference (pp)	95% interval
1 h	Pre-trigger adjusted	62	58.1 / 38.5	+18.0	[+5.7, +30.4]
1 h	Route decomposition	62	58.1 / 38.5	+13.0	[+0.6, +25.5]
6 h	Pre-trigger adjusted	86	57.0 / 38.1	+18.5	[+7.4, +29.5]
6 h	Route decomposition	86	57.0 / 38.1	+12.9	[+1.4, +24.3]
24 h	Pre-trigger adjusted	104	55.8 / 37.9	+18.6	[+8.4, +28.7]
24 h	Route decomposition	104	55.8 / 37.9	+13.1	[+2.4, +23.7]
48 h	Pre-trigger adjusted	109	55.0 / 37.9	+17.3	[+7.3, +27.4]
48 h	Route decomposition	109	55.0 / 37.9	+11.6	[+0.9, +22.3]
48 h	Repository fixed effects	109	55.0 / 37.9	+18.6	[+3.7, +33.5]
48 h	Leave one ordered product pair out	at least 70	–	+14.7 to +21.8	lower bounds +4.1 to +9.9
48 h	Specificity: exact edge vs non-exact discussion	109 of 615	55.0 / 41.7	+12.5	[+1.4, +23.5]
48 h	Specificity: overlap weighting	109 of 615	–	+11.5	[+0.3, +22.8]
48 h	Specificity: repository fixed effects	109 of 615	–	+15.6	[-4.4, +35.6]; 37 varying repos

Note: The main models use 1,067 inline-trigger PRs in 469 repositories that were open at hour 48 and had a complete 30-day horizon. Specificity rows compare 109 exact-edge PRs with 506 PRs that had public discussion but no exact edge. This post-trigger discussion control is a structural falsification check, not a causal or semantic-resolution estimate.

Table 15 Four-state contextual response gradient at the 48-hour landmark.

Contrast	Compared group	Reference group	Difference (pp)	95% interval
Exact edge vs no visible activity	109 PRs; 55.0%	426 PRs; 31.0%	+24.2	[+13.5, +34.9]
Non-exact discussion vs no visible activity	506 PRs; 41.7%	426 PRs; 31.0%	+10.0	[+2.8, +17.3]
Movement only vs no visible activity	26 PRs; 76.9%	426 PRs; 31.0%	+38.8	[+19.8, +57.7]
Exact edge vs non-exact discussion	109 PRs; 55.0%	506 PRs; 41.7%	+14.2	[+3.7, +24.6]

Note: One model over the full 1,067-PR landmark cohort in 469 repositories places all four post-trigger states against the no-visible-activity reference, using the same pre-trigger controls as the primary model. The four states are mutually exclusive, so the groups sum to the cohort. The movement-only group holds 26 PRs, so its interval is wide and its point estimate is not a ranking. The active-discussion subset remains the primary specificity contrast because it does not compare an active PR with a silent one.

Table 16 Forty-eight-hour ownership routes and later integration.

Route	PRs	Later merge (%)	Median first action (h)	Adjusted difference (pp)	Repository-cluster 95% interval
Automation, no later user event	314	40.1	0.09	Reference	–
Automation then user	162	54.9	0.06	+12.9	[+3.0, +22.7]
User first	467	50.1	0.51	+6.1	[-1.7, +13.9]
Movement only	55	58.2	2.00	+11.6	[-6.7, +30.0]
Other activity	76	39.5	0.06	-1.5	[-14.5, +11.6]
No observed action	659	32.2	–	-9.2	[-15.9, -2.5]

Note: The reference is automation with no later user event. Adjustment uses only pre-trigger activity and trigger context. Routes are observed markers, not assigned treatments.

Table 17: Sensitivity of the addressed-edge later-merge association to unmeasured structure.

Check	Reply window	Estimate (pp)	95% interval or reference band	Sensitivity summary
Unmeasured confounding	1 h	+18.0	[+5.7, +30.4]	risk ratio 1.47; E-value 2.30, limit 1.56
Unmeasured confounding	6 h	+18.5	[+7.4, +29.5]	risk ratio 1.48; E-value 2.33, limit 1.68
Unmeasured confounding	24 h	+18.6	[+8.4, +28.7]	risk ratio 1.49; E-value 2.34, limit 1.74
Unmeasured confounding	48 h	+17.3	[+7.3, +27.4]	risk ratio 1.46; E-value 2.27, limit 1.67
Tipping point for one binary factor	48 h	–	–	a factor 20 pp more common among edge PRs must itself carry 86.7 pp of later merge to remove the point estimate, and 36.3 pp to remove the interval
Tipping point for one binary factor	48 h	–	–	a factor 40 pp more common among edge PRs must itself carry 43.4 pp of later merge to remove the point estimate, and 18.1 pp to remove the interval
Negative control: pre-trigger decisive review	1 h	-0.1	[-5.7, +5.5]	interval covers the null
Negative control: pre-trigger branch movement	1 h	+0.3	[-7.6, +8.2]	interval covers the null
Negative control: pre-trigger user event	1 h	+16.7	[+4.9, +28.4]	interval excludes the null: residual pre-trigger confounding
Negative control: pre-trigger decisive review	6 h	-0.5	[-4.8, +3.8]	interval covers the null
Negative control: pre-trigger branch movement	6 h	+0.5	[-6.5, +7.4]	interval covers the null
Negative control: pre-trigger user event	6 h	+5.8	[-3.9, +15.4]	interval covers the null
Negative control: pre-trigger decisive review	24 h	-1.1	[-4.7, +2.5]	interval covers the null
Negative control: pre-trigger branch movement	24 h	+0.3	[-5.6, +6.2]	interval covers the null

Sensitivity of the addressed-edge later-merge association to unmeasured structure. (continued)

Check	Reply window	Estimate (pp)	95% interval or reference band	Sensitivity summary
Negative control: pre-trigger user event	24 h	+8.3	[-0.7, +17.3]	interval covers the null
Negative control: pre-trigger decisive review	48 h	-0.3	[-4.0, +3.3]	interval covers the null
Negative control: pre-trigger branch movement	48 h	+1.3	[-4.3, +6.9]	interval covers the null
Negative control: pre-trigger user event	48 h	+7.0	[-1.7, +15.7]	interval covers the null
Randomisation inference (unconditional)	1 h	+18.0	[+2.0, +16.8]	two-sided p = 0.010 over 2,000 within-repository permutations across 32 repositories
Randomisation inference (unconditional)	6 h	+18.5	[+1.3, +14.5]	two-sided p = 0.001 over 2,000 within-repository permutations across 37 repositories
Randomisation inference (unconditional)	24 h	+18.6	[+2.7, +14.4]	two-sided p = 0.001 over 2,000 within-repository permutations across 46 repositories
Randomisation inference (unconditional)	48 h	+17.3	[+2.8, +14.4]	two-sided p = 0.001 over 2,000 within-repository permutations across 46 repositories

Note: All rows use the 1,067-PR inline-trigger landmark cohort and the pre-trigger adjusted specification. E-values are computed on an approximate risk-ratio scale from the adjusted risk difference and the unexposed later-merge rate; they state the minimum association an unmeasured factor would need with both the exact edge and later merge, beyond the measured controls. Negative-control outcomes were complete before the trigger, so the exposure cannot produce them and a non-null estimate signals residual confounding rather than an effect. The randomisation rows are different from the others in two ways. They are the unconditional test, which permutes across the whole cohort, and their bracketed pair is the 2.5th to 97.5th percentile of the permuted reference distribution, not a confidence interval for the estimate. That reference distribution is not centred on zero because 423 of 469 repositories contain no exposure variation. The centred conditional version, which is the one quoted in the article, is in the exposure-scope table. None of these checks identifies a causal effect.

Table 18: Pre-trigger covariate balance and propensity overlap for the exact-edge contrast.

Reply window	Quantity	Edge mean	No-edge mean	Standardized difference
1 h	Log trigger age in hours since PR creation	1.117	1.228	-0.059
1 h	Log pre-trigger interaction events	1.043	0.910	+0.158
1 h	Pre-trigger user-account events	1.065	0.830	+0.096
1 h	Pre-trigger bot events	2.177	2.277	-0.018
1 h	Pre-trigger decisive reviews	0.048	0.066	-0.051
1 h	Pre-trigger branch movements	0.210	0.216	-0.006
6 h	Log trigger age in hours since PR creation	1.075	1.234	-0.085
6 h	Log pre-trigger interaction events	0.943	0.916	+0.032
6 h	Pre-trigger user-account events	0.767	0.850	-0.035
6 h	Pre-trigger bot events	1.953	2.299	-0.065
6 h	Pre-trigger decisive reviews	0.035	0.067	-0.097
6 h	Pre-trigger branch movements	0.174	0.219	-0.045
24 h	Log trigger age in hours since PR creation	1.024	1.243	-0.118
24 h	Log pre-trigger interaction events	0.960	0.914	+0.057
24 h	Pre-trigger user-account events	0.788	0.849	-0.025
24 h	Pre-trigger bot events	1.904	2.310	-0.078
24 h	Pre-trigger decisive reviews	0.029	0.069	-0.119
24 h	Pre-trigger branch movements	0.192	0.218	-0.025
48 h	Log trigger age in hours since PR creation	1.110	1.234	-0.066
48 h	Log pre-trigger interaction events	0.970	0.912	+0.070
48 h	Pre-trigger user-account events	0.780	0.851	-0.030
48 h	Pre-trigger bot events	1.963	2.306	-0.066
48 h	Pre-trigger decisive reviews	0.046	0.067	-0.059
48 h	Pre-trigger branch movements	0.229	0.214	+0.015
48 h	Propensity score at the 0 percentile	0.040	0.000	–
48 h	Propensity score at the 5 percentile	0.083	0.040	–

Pre-trigger covariate balance and propensity overlap for the exact-edge contrast. (continued)

Reply window	Quantity	Edge mean	No-edge mean	Standardized difference
48 h	Propensity score at the 25 percentile	0.155	0.089	–
48 h	Propensity score at the 50 percentile	0.218	0.142	–
48 h	Propensity score at the 75 percentile	0.299	0.215	–
48 h	Propensity score at the 95 percentile	0.690	0.307	–
48 h	Propensity score at the 100 percentile	0.737	0.759	–

Note: The first block reports the six measured pre-trigger controls in the 1,067-PR landmark cohort. A standardized mean difference is the group difference divided by the pooled standard deviation. Every value is below 0.16 in absolute size. The last block reports the propensity-score distribution of the 109 exact-edge PRs and the 506 non-exact-discussion PRs, so a reader can see where the two groups overlap and where they do not. Balance on measured variables says nothing about unmeasured structure such as task difficulty.

Table 19: Scope of the addressed-edge exposure and of the landmark cohort.

Aspect	Item	Size	Value
Cohort restriction	Cross-product inline triggers with a 30-day horizon	3,942	100.0% of cross-product inline triggers
Cohort restriction	Closed at or before hour 48	2,875	72.9% of cross-product inline triggers
Cohort restriction	Merged at or before hour 48	2,528	64.1% of cross-product inline triggers
Cohort restriction	Still open at hour 48	1,067	27.1% of cross-product inline triggers
Who writes the edge	PR author's own user account	76 events	59.4% of exposure events on 74 PRs
Who writes the edge	Another user account	29 events	22.7% of exposure events on 26 PRs
Who writes the edge	The triggering product itself	13 events	10.2% of exposure events on 13 PRs
Who writes the edge	An unmapped bot	7 events	5.5% of exposure events on 3 PRs
Who writes the edge	The PR author's product	2 events	1.6% of exposure events on 2 PRs
Who writes the edge	A different mapped product	1 event	0.8% of exposure events on 1 PR
Exposure definition	Any exact parent reply (primary)	109	+17.3 pp [+7.3, +27.4]
Exposure definition	Exact reply, excluding the triggering product's own reply	99	+15.9 pp [+5.4, +26.3]
Exposure definition	Exact reply written by a user account	97	+17.0 pp [+6.4, +27.6]
Exposure definition	Exact reply written by a different mapped product	3	not modelled: too few exposed PRs
Within-repository randomisation	Repositories that can be re-randomised, with repository fixed effects	321 PRs in 46 repos	observed +16.4 pp; reference mean -0.1 pp over 2,000 permutations; two-sided p = 0.007

Note: The exposure rule requires a reply whose parent identifier is the trigger's own identifier; it does not require another product to write it. The composition rows report who actually wrote the 128 exposure events on the 109 exposed PRs. The exposure-definition rows refit the primary pre-trigger adjusted model on the same 1,067-PR cohort under stricter definitions. The cohort-restriction rows show that the landmark cohort is the slower-resolving remainder of the cross-product inline-trigger population. The randomisation row uses only repositories containing both exposed and unexposed PRs, so its reference distribution is centred; an unconditional permutation of a model without repository fixed effects is not, because the between-repository part of the coefficient is invariant to a within-repository permutation.

Table 20: Extensions to RQ3: the whole population over time, and who writes the edge.

Analysis	Size	Estimate	95% interval
Whole population: time only	3,940 PRs, 13,719 periods	1.73	[1.39, 2.15]; p = 1.1e-06
Whole population: products and month	3,940 PRs, 13,719 periods	1.74	[1.38, 2.18]; p = 2.5e-06
Whole population: full pre-trigger controls	3,940 PRs, 13,719 periods	1.74	[1.38, 2.18]; p = 2.3e-06
Edge written by a prior reviewer of this repository	64 PRs, raw 71.9%	+31.3 pp	[+20.0, +42.5]
Edge written by an account new to this repository	32 PRs, raw 28.1%	-9.5 pp	[-26.1, +7.1]
Edge written by automation	13 PRs, raw 38.5%	+16.0 pp	[-12.4, +44.3]
Prior reviewer minus newcomer: primary	pre-trigger adjusted	+40.7 pp	[+22.4, +59.1]
Prior reviewer minus newcomer: repository fixed effects	within-repository comparison only	+8.7 pp	[-16.8, +34.2]
Prior reviewer minus newcomer: leave one ordered product pair out	21 refits	+40.7 pp	[+37.1, +42.8]
Prior reviewer minus newcomer: leave one exposed repository out	76 refits	+40.4 pp	[+38.1, +43.3]

Note: The whole-population rows drop the hour-48 restriction. Every cross-product inline-trigger PR with a complete 30-day horizon is followed from its trigger, follow-up is split into eleven intervals, and the exact addressed edge enters as a time-varying covariate, so a PR contributes unexposed periods until its own first exact reply. The estimate is an odds ratio for merging in the next interval, from a pooled logistic model with interval indicators and repository-clustered standard errors. The edge-class rows split the 109 exposed PRs of the landmark cohort by the account that wrote the first exact reply, using the same strict prior-history rule as the RQ2 analysis, against the no-edge reference. The final block repeats that split under repository fixed effects and under leave-one-out exclusions. The fixed-effect row is the decisive one: the gap between a prior reviewer and a newcomer is largely a difference between repositories, not within them, so these rows describe where the signal sits and do not identify a mechanism.

Table 21 Structural same-locus review overlap and frozen semantic gates.

Quantity or gate	Observed	Interpretation or status
Eligible PRs with two or more mapped reviewer products	886	Context
Canonical same-snapshot, same-locus pairs	167	Structural population
Pull requests with at least one locus	159	17.9% of eligible PRs
Repositories / product pairs	98 / 8	Support
Median gap	1.37 min	Timing
Pairs within five minutes	86.8%	cluster interval [80.9, 92.6]%
Open-state pairs within five minutes	85.7%	133 loci
Non-dominant-pair loci within five minutes	82.8%	58 loci
Exact normalized comment-body duplicates	0	Not a semantic comparison
Largest repository share	5.4%	PASS
Largest product-pair share	65.3%	FAIL
Blinded packet coverage	167	Complete population
Dual-coder agreement	Pending	Need kappa at least 0.70
Semantic relation labels	Pending	No duplication or complementarity claim

Note: Structural co-location and timing do not establish semantic duplication, contradiction, correctness, repair, or coordination. The product-pair generality gate fails because the largest pair exceeds 50 percent.

Table 22: External evidence ladder after schema, overlap, and construct checks.

Source	Native unit	Exact edge	Role after audit
GH Archive plus GitHub REST enrichment	One public GitHub event; enriched review comment or PR record	YES after REST	Future disjoint time-window replication after REST enrichment; not used in the present estimates
SWE-Review-Chat	One PR with ordered general discussion, reviews, inline comments and nested thread replies	PARTIAL	Full-corpus topology audit; no independent exact-edge cohort survived overlap exclusion
AI-to-AI Code Reviews closed-loop cohorts	One PR-reviewer-product pair	NO	Product-pair attribution and trigger-time sensitivity on overlapping public PRs
SWE-PRBench	One merged PR plus selected human-written initiating comments; synthetic comment IDs are scoped only within task	NO	Semantic codebook and initiating-feedback audit only; reply topology is not preserved
SWE-Review-Traj	One decision-correct agentic reviewer trajectory over a generated candidate PR	NO	Controlled generate–review–revise contrast only; not a field replication
GitHub Agentic PR Dataset	One PR, commit, or changed file depending on table	NO	Excluded: adds PRs and diffs but no review-response graph
Trace Commons Agent Traces	One voluntarily donated coding-agent session	NO	Excluded: opt-in private sessions with no stable public PR edge or outcome

Note: The registry contains 15 screened candidates. Sources are not pooled merely to add rows. A source must preserve the PR, event time, actor, review batch, exact reply parent, and later-state horizon before it can reproduce the declared topology.

Table 23 Fail-closed exact-edge replication audit in SWE-Review-Chat.

Stage	PRs or records	Meaning
Public PR rows scanned	1,082,529	Available corpus
PRs whose author maps by the frozen exact alias list	5,250	Product-aware author support
PRs with a cross-product inline parent	38	Candidate trigger support
PRs where such a parent has a nested reply	7	Seven candidate PRs before overlap removal
Nested children before overlap removal	18 / 1 / 0	Unmapped user / same product / different product
Candidate PRs outside the AIDev full corpus	0	No independent cohort remains
REST-validated 48-hour landmark PRs	0	Replication gate fails

Note: All seven pre-exclusion candidate PRs already occur in the AIDev full corpus. No GitHub comment or PR body was exported. Zero disjoint support is a failed replication gate, not evidence that exact public connections never occur.

Table 24 Independent-packaging sensitivity for product attribution and trigger anchoring.

Check	PRs	Observed	Use
PR overlap	123	11.5% of landmark cohort	Coverage only
Exact author–reviewer pair agrees	119	96.7% of overlapping PRs	Supports product attribution
Trigger time agrees within five minutes	104	99.0%	Supports event anchoring
Exact-edge exposed PRs	9	7.6% of exact-pair overlap	Too few for replication
Raw later-merge difference	119	+14.6 pp	[-18.6, +47.9]
Pre-trigger-adjusted difference	119	+11.1 pp	[-23.9, +46.1]

Note: The external cohort was released independently but observes overlapping public GitHub PRs. Its 119-PR exact-pair overlap contains only nine exposed PRs. The merge rows are therefore an appendix sensitivity, not an independent outcome replication or a causal estimate.

Table 25: Load-bearing data-quality, temporal, and leakage checks.

Gate	Status	Observed
Full-corpus PR identifiers are unique	PASS	7,685,281
Follow-up event PR identifiers resolve	PASS	0 orphan IDs
Events are strictly after the trigger	PASS	0 violations
Events stay inside the seven-day window	PASS	0 violations
Later review batches are de-duplicated	PASS	0 repeated batches
Exact duplicate event surplus removed	PASS	9 rows
First state is invariant to exact de-duplication	PASS	0 changed states
Mapped-first anchor reconciles	PASS	924 PRs
Landmark cohort has one row per PR	PASS	1,067 PRs
Direct-reply parent identifiers equal the trigger	PASS	128 reply events
All positive outcomes occur after hour 48	PASS	423 outcomes
All outcomes stay within the 30-day horizon	PASS	True
Pre-trigger controls are strictly pre-trigger	PASS	3,323 rows
Prior-history matches exclude focal and future rows	PASS	221,058 valid rows
Review-request target account coverage	LIMIT	0.0%

Note: PASS means that the frozen automated check met its declared contract. LIMIT identifies missing measurement coverage rather than a failed computation.

Table 26: Variable dictionary for the 31-column landmark cohort.

Column	Type	Available when	Definition
pr_id	Int64	At or before the trigger	PR identifier; analysis grain is one PR's first cross-product inline trigger.
repo_id	Int64	At or before the trigger	Repository cluster and fixed-effect identifier.
trigger_dt	Datetime (UTC)	At or before the trigger	Timestamp of the first recognized cross-product feedback event on the PR.
trigger_event_id	Int64	At or before the trigger	Raw inline-comment id of the exact trigger.
trigger_review_id	Int64	At or before the trigger	Review-batch id containing the trigger.
author_agent	String	At or before the trigger	Mapped product attributed as PR author.
trigger_reviewer_agent	String	At or before the trigger	Exactly mapped product that authored the trigger.
ordered_product_pair	String	At or before the trigger	Directed author product → reviewer product.
trigger_month	String	At or before the trigger	Internal timing or audit field.
outcome_landmark_dt	Datetime (UTC)	At or before the trigger	Trigger timestamp plus 48 hours; every PR is still open here.
merged_from_48h_to_30d	Boolean	Outcome window, after hour 48	Observed merge strictly after the landmark and no later than trigger + 30 days.
first_exact_reply_hours	Float64	After the trigger, by hour 48	Hours to the earliest inline reply whose in_reply_to_id equals trigger_event_id.
first_exact_reply_dt	Datetime (UTC)	After the trigger, by hour 48	Timestamp of that earliest exact-parent reply.
exact_reply_events_48h	UInt32	After the trigger, by hour 48	Number of exact-parent inline replies by the 48-hour landmark.
exact_parent_reply_by_1h	Boolean	After the trigger, by hour 48	True if any inline reply with in_reply_to_id exactly equal to trigger_event_id occurs within 1 hour; structural edge only.
exact_parent_reply_by_6h	Boolean	After the trigger, by hour 48	True if any inline reply with in_reply_to_id exactly equal to trigger_event_id occurs within 6 hours; structural edge only.
exact_parent_reply_by_24h	Boolean	After the trigger, by hour 48	True if any inline reply with in_reply_to_id exactly equal to trigger_event_id occurs within 24 hours; structural edge only.

Variable dictionary for the 31-column landmark cohort. (continued)

Column	Type	Available when	Definition
exact_parent_reply_by_48h	Boolean	After the trigger, by hour 48	True if any inline reply with in_reply_to_id exactly equal to trigger_event_id occurs within 48 hours; structural edge only.
ownership_route_48h	String	After the trigger, by hour 48	Post-trigger public route observed by 48 hours; used only in secondary Model B.
pre_events	UInt32	At or before the trigger	Count of valid submitted-review, PR-comment, and inline-comment rows strictly before trigger.
pre_user_events	UInt32	At or before the trigger	Pretrigger interaction rows whose GitHub user_type is User.
pre_bot_events	UInt32	At or before the trigger	Pretrigger interaction rows whose GitHub user_type is Bot.
pre_decisive_reviews	UInt32	At or before the trigger	Pretrigger submitted reviews with APPROVED or CHANGES_REQUESTED state.
pre_force_pushes	UInt32	At or before the trigger	Timestamped force-push timeline rows strictly before trigger.
log1p_trigger_age_hours	Float64	At or before the trigger	log(1 + hours from PR creation to trigger).
log1p_pre_events	Float64	At or before the trigger	log(1 + pre_events).
pr_created_dt	Datetime (UTC)	At or before the trigger	Internal timing or audit field.
last_pre_event_dt	Datetime (UTC)	At or before the trigger	Internal timing or audit field.
last_pre_force_push_dt	Datetime (UTC)	At or before the trigger	Internal timing or audit field.
merged_dt	Datetime (UTC)	Outcome window, after hour 48	Internal timing or audit field.
closed_dt	Datetime (UTC)	At or before the trigger	Internal timing or audit field.

Note: The availability column is the leakage tag stored with the cohort itself. Only columns available at or before the trigger enter the primary model. Columns available after the trigger enter the secondary route decomposition and the exposure definition. Columns tagged as outcome are read only after hour 48. Datetime columns are microsecond UTC timestamps.

Table 27: Resampling and permutation settings by analysis.

Analysis	Resampling unit	Draws	Seed	Interval method
Matched cross/same visibility contrast	Whole repositories	10,000	20260826	Percentile band on the pair-weighted difference
Matched contrast, pair-level check	Matched pairs	10,000	20260826	Percentile band
Burst first-state shares	Whole repositories	5,000	20260826 plus the threshold in minutes	Percentile band on each state share
Burst mapped-product retention	Whole repositories	5,000	20260826 plus 1,000 plus the threshold	Percentile band on the difference
Deep next-owner shares	Whole repositories	5,000	20260826 plus 100 plus the washout	Percentile band
Same-product continuation placebo	Whole repositories	5,000	20260826 plus 200 plus the washout	Percentile band on the difference
Within-PR product-order placebo	Event order inside one PR	500 permutations	20260826 plus 400	Permutation mean and percentile band
Automation-episode order placebo	Event order inside one PR	500 permutations	20260826 plus 300 plus the gap and state index	Permutation mean and percentile band
Ownership persistence contrasts	Whole repositories	5,000	20260826 plus a fixed metric offset	Percentile band
Same-locus timing share	Whole repositories	10,000	20260826	Percentile band
Randomisation inference, unconditional	Exposure labels inside each repository	2,000 permutations	20260826	Two-sided permutation p and percentile band
Randomisation inference, conditional	Exposure labels inside each varying repository	2,000 permutations	20260826	Two-sided permutation p against a centred reference
All outcome regressions	None	–	–	Cluster-robust normal approximation, not resampling

Note: Draw counts are not uniform across the analyses, so they are listed rather than summarized. Every seed is fixed, and offsets are added so that separate quantities inside one script do not reuse one draw sequence. A percentile band is the 2.5th and 97.5th percentile of the draw distribution. The last row records that the regression intervals come from a clustered normal approximation, which is why the exact-edge estimate is also checked by permutation.

Table 28: Ordered reproduction contract with frozen row counts.

Step	Command	Produces	Expected rows
1	`uv sync`	Resolves and installs the pinned dependency graph	–
2	`run_cross_agent_review_exploration.py`	Cross-product trigger cohort and seven-day response chains	–
3	`run_response_ownership_analysis.py`	First response owner and 48-hour ownership routes	6 rows
4	`run_response_ownership_robustness.py`	Leave-one-out ranges for the ownership descriptives	10 rows
5	`run_coordination_topology_analysis.py`	Participation funnel, exact-author matched contrasts, route contrasts	10 rows
6	`run_burst_collapsed_topology.py`	Burst-threshold first states, tie diagnostics, ordering robustness	25 rows
7	`run_deep_coordination_transitions.py`	Post-burst next owners, episode sessionization, order placebos	24 rows
8	`run_legacy_extension_ownership_persistence.py`	Exact-owner and layer persistence falsification	3 rows
9	`run_human_memory_bridge_analysis.py`	Strict prior same-repository review history	3 rows
10	`run_addressed_edge_landmark_analysis.py`	Landmark cohort, schema, balance, primary exact-edge models	8 rows
11	`run_addressed_edge_specificity_analysis.py`	Discussion controls, overlap weighting, four-state gradient	6 rows
12	`run_addressed_edge_confounding_sensitivity.py`	E-values, tipping grid, negative controls, permutation test	4 rows
13	`run_addressed_edge_scope_audit.py`	Exposure composition, stricter definitions, conditional permutation	6 rows
14	`run_rq3_extensions.py`	Whole-population time-varying hazard, and the edge split by who wrote it	3 rows
15	`prepare_review_collision_audit.py`	Blinded structural same-locus coder packets	8 rows
16	`run_collision_descriptive_extension.py`	Descriptive timing and concentration for the same-locus population	5 rows
17	`generate_technical_appendix_tables.py`	Every table in this appendix	–
18	`validate_response_ownership_outputs.py`	Re-checks the ownership products against their frozen contracts	–
19	`validate_coordination_extension_outputs.py`	Re-checks the topology and landmark products	–
20	`visualize_manuscript_figures.py`	The five article figures and the two appendix figures	–

Ordered reproduction contract with frozen row counts. (continued)

Step	Command	Produces	Expected rows
21	`uv run --with pytest python -m pytest -q`	Automated test suite	–

Note: The order is read from the repository README, so this table cannot drift from the documented sequence. The expected-row column names one frozen product per step where a stable count exists; a dash means the step writes figures, validation logs, or binary tables instead. A rerun that changes a listed count is a signal to compare inputs, not to accept the new number.

Table 29: Experiment and claim disposition after robustness and construct checks.

Experiment or claim	Status	Reason
Product co-presence as collaboration	REJECT	Two products on one PR do not establish an addressed or sequential edge
Burst-collapsed next public owner	MAIN	Ordering remains stable across thresholds and leave-one-group checks
Cross-product versus same-product visibility	MAIN	Exact-author matched contrast remains negative with repository resampling
Prior same-repository review history of user-account bridge	MAIN	Strict pre-trigger different-PR history is common and concentration checks pass
Experienced accounts are preferentially selected first	REJECT	The all-responder comparison differs little and channel or account dependence remains
Hybrid automation-then-user route and later merge	SECONDARY	Positive pretrigger-adjusted association triangulates the exact-edge result in a broader fixed-landmark cohort
Exact addressed reply and later merge	MAIN	Pre-landmark association is stable across reply windows and product-pair exclusions
Exact edge versus non-exact public discussion	MAIN	The association remains positive inside PRs with visible discussion and under overlap weighting; repository fixed effects are imprecise
Special same-product temporal persistence	APPENDIX	Observed ordering is reproduced by the within-PR product-composition placebo
Repeated automation naturally escalates to a user	REJECT	The apparent dose curve is reproduced by within-PR order permutation
One-way user takeover after a rapid burst	REJECT	The next visible owner bounces in both directions and same-layer persistence is similar for user-first and product-first states
Prior-review history causes later merge	REJECT	The positive pooled association does not survive strong within-repository comparisons
Semantic duplication or complementarity at the same locus	PENDING	Structural cohort is ready but dual coding and product-pair generality gates remain open
Intentional routing to the triggering reviewer product	REJECT	Review-request assignee coverage is zero
Second-brand adoption expands repository task breadth	REJECT	The pattern fails longer-history and rarefaction sensitivity checks
Exact-file succession is a true task handoff	REJECT	File reuse alone fails the task-identity audit

Experiment and claim disposition after robustness and construct checks. (continued)

Experiment or claim	Status	Reason
Independent CodAGE product-attribution agreement	APPENDIX	Exact product pairs and trigger times agree in the overlap but only nine exposed PRs remain for an outcome check
SWE-Review-Chat exact-edge outcome replication	REJECT	All seven PRs whose exact-alias cross-product inline parent has a nested reply are already in the complete AIDev backbone leaving zero disjoint REST or landmark rows
Transferred actionability text classifier	REJECT	Leave-one-source-out discrimination is weak for human review and one AI-review tool so the proxy is not stable enough for AIDev claims
Unmeasured-confounding bounds for the addressed edge	MAIN	E-value 2.27 and an explicit tipping grid state the minimum strength a confounder would need instead of only warning that one could exist
Pre-trigger placebo outcomes for the addressed edge	MAIN	Eleven of twelve models cover the null at the same specification and the exposure cannot cause an outcome completed before the trigger
One-hour reply window as the primary exposure	REJECT	The pre-trigger user-activity placebo excludes the null at one hour so a very fast exact reply is selected by PRs that already had a person acting
Repository-stratified randomisation inference	SECONDARY	A design-based reference distribution confirms the clustered interval is not an artefact of 109 exposed PRs
Addressed edge as an agent-to-agent relay	REJECT	Only three of 128 exposure events are written by a different mapped product so the edge is a human acknowledgement in this cohort
Addressed edge under stricter exposure definitions	MAIN	Dropping the triggering product's own replies and requiring a user-account reply both keep the estimate positive with overlapping intervals
Landmark cohort as a general population estimate	REJECT	The cohort is the 27 percent of cross-product inline triggers still open at hour 48 so the estimate describes slower-resolving PRs only
Conditional within-repository randomisation test	MAIN	Restricting to repositories that can be re-randomised and adding repository fixed effects centres the reference distribution and gives a two-sided p of 0.007

Experiment and claim disposition after robustness and construct checks. (continued)

Experiment or claim	Status	Reason
Whole-population time-varying edge model	MAIN	Dropping the hour-48 restriction and letting the edge switch on in time leaves the odds of merging about 1.7 times higher across every control set
Repository familiarity as the mechanism behind the edge	REJECT	The prior-reviewer minus newcomer gap is large overall but shrinks to near zero with an interval spanning it once the comparison is held inside one repository
Edge-class description by who wrote the reply	APPENDIX	Informative about where the association sits but confounded with the repository so it cannot support a mechanism claim

Note: MAIN enters the article's core story; SECONDARY supports that story; APPENDIX is informative but not a headline; REJECT is retained as a falsification result; PENDING requires evidence not yet available.

4 Identity registry and trigger construction

Product attribution uses an allowlist of exact vendor-controlled GitHub identities. Similar names and unknown bots are not guessed. The complete registry is in Table 3. The unit for the main cohort is one PR with its first mapped feedback event written by a product different from the agent label attributed to the PR author.

Eligible feedback sources are a submitted review, an inline review comment, or a PR comment. The event must be after PR creation and no later than PR closure. The common observation boundary is 15 April 2026 at 00:00 UTC. PR creation is cut off on 31 March 2026, which leaves time to observe follow-up events.

For an inline trigger, a direct reply must point to the trigger's exact comment identifier. A reply elsewhere on the PR does not count. If the inline trigger belongs to a submitted review, the enclosing review is the same batch and is not counted as a later review. A later review needs a different review-batch identifier. These rules prevent one API batch from answering itself.

Several short terms carry weight across all three research questions, so Table 4 states each one as the rule the code applies. It covers a decisive review, branch movement, the residual untyped class, the five mutually exclusive actor states, the seven response actor roles, and the automation episode with its five-minute gap parameter. Each definition was read back from the analysis code rather than restated from the article. Every entry is a rule for labelling a public event. The table does not say how often each label occurs, and it does not say what any labelled event meant to the people or products involved.

5 Cohorts and denominators

Table 5 gives the full cohort flow. The seven-day topology cohort requires a complete observation window and ends earlier if the PR closes. Exact-parent replies are possible only for inline triggers; therefore the article reports both the full-cohort and inline-eligible denominators.

The outcome cohort uses a different rule. It first observes public activity for 48 hours after the trigger, retains PRs still open at hour 48, and then observes merge from hour 48 through day 30. It also requires a complete 30-day outcome window. This landmark keeps the route definition before the outcome window.

6 Estimation specifications

Every outcome model in this paper is a linear probability model fitted by ordinary least squares, with standard errors clustered on the repository. Table 6 lists each named specification with its outcome, its exposure term, its estimator, its standard-error type, its denominators, and its full model formula. Most formulas are stored with the analysis products themselves, so the table cannot drift from the fits it describes. The two ownership-route rows and the two prior-history rows are read back from the analysis code, which does not write a formula column. Reply-window variants at 1, 6, and 24 hours refit the same formula with the exposure term for that window. The table shows what was fitted and on how many PRs. It does not show that any of these fits identifies a causal effect, and a linear probability model can predict outside the unit interval.

7 RQ1 details: burst collapse and relay topology

The rapid-burst analysis repeats the first-owner classification after excluding events in the first 0, 1, 5, 10, or 30 minutes. The threshold is a sensitivity parameter, not a recovered private session boundary. Actor states are mutually exclusive: user account, mapped product, other bot, branch movement or untyped activity, and no later action. Tied timestamps use a fixed priority order and are separately checked. Table 7 gives all thresholds and repository-cluster intervals.

The priority order is user account, then mapped product, then other bot, then branch movement or untyped activity. It is applied whenever two or more events share the earliest post-burst timestamp, so the assigned state does not depend on input row order. Table 8 reports how often that rule is actually used. Tied first timestamps are common, because many GitHub events are stamped to the whole second. A tie changes the assigned state only when the tied events fall in different states. Exactly one such mixed-state tie occurs, at the zero-minute threshold, and none occurs at any positive threshold. This matters because the order favours the user-account state, and the RQ1 ordering claim rests on that state. The diagnostic shows the rule is almost never the deciding factor, not that it would be harmless if it were.

The article and this appendix both state that the ordering survives every leave-repository-out and leave-product-pair-out check. Table 9 gives the numbers behind that sentence. The first block drops one repository at a time, across 1,416 repositories, and then one ordered author–reviewer product pair at a time, across 25 pairs. Two gates are checked in every exclusion: the user-account share stays above the mapped-product share, and no later action within seven days stays the largest state. Both gates hold in all exclusions at all five thresholds. At the primary five-minute threshold the user-minus-mapped gap stays between 16.5 and 19.9 points. The second block reports the leave-one-out share range for each state at that threshold, where the largest single shift is 3.42 points. These checks bound concentration in one repository or one product pair. They do not address selection into the cohort itself.

The deep transition test starts with PRs whose first state after the five-minute burst is a mapped product. It then applies another five-minute washout and asks who owns the next distinct state. A within-PR random-order benchmark preserves each PR's later product composition while breaking temporal order. This test guards against calling a dominant product mix a special persistence mechanism. Table 10 gives the observed transitions and placebo.

A second exploratory test sessionizes public automation episodes and asks whether repeated automation naturally escalates to a user account. The apparent drop in user-next probability is reproduced by within-PR order permutation. The paper therefore rejects an escalation-dose claim; the full falsification is reported in Table 11.

8 RQ2 details: boundary matching and prior public history

The cross/same comparison pairs triggers with the same repository, exact PR author account, author product, trigger source, and calendar month. Pairing uses nearest trigger time without replacement. The main interval resamples whole repositories; a seven-day time caliper is a sensitivity analysis. Table 12 reports all measured outcomes rather than only the main visibility contrast.

For the first user-account mediator, prior history requires the same account, same repository, a different PR, and a submitted-review timestamp strictly before the trigger. The same rule is applied to the first later decisive reviewer. Table 13 gives role, channel, depth, and comparison populations. The all-responder comparison is important: prior history is common throughout the response layer, so the analysis does not claim that experienced accounts are preferentially selected to respond first.

A review-request event is visible before many triggers, but the exported event does not identify the requested account. Assignee, message, and label coverage is zero in the valid request rows. The paper therefore rejects the claim that the triggering product was deliberately routed or requested.

9 RQ3 details: connected signals and later integration

Two pre-landmark topology signals are related to the later-state question. The first is an exact addressed reply within the initial 48 hours among PRs with an inline trigger. The second is the mutually exclusive ownership route in the full landmark cohort. The route categories are user account first, automation then user, automation with no user event, movement only, other activity, or no observed action.

For the exact-edge model, the primary specification adjusts only for variables known before the trigger: author product, reviewer product, month, trigger age, and prior user, bot, decisive-review, force-push, and total public activity. A secondary decomposition also includes the 48-hour ownership route. The reply window is varied over 1, 6, 24, and 48 hours. Repository fixed effects and leave-one-product-pair-out fits are sensitivity analyses.

A specificity control removes PRs with no public discussion by hour 48. It compares an exact parent edge with later reviews or PR comments that do not point to the trigger. An overlap-weighted fit is a second measured-balance check. This control asks whether the edge differs from generic visible discussion; it still cannot tell whether either discussion understood or resolved the review point.

A fourth, wider view fits all four post-trigger states in one model on the full 1,067-PR cohort, against a no-visible-activity reference. Table 15 reports it with its denominators. Relative to the 426 PRs with no visible activity by hour 48, later merge is +24.2 points higher for the 109 exact-edge PRs and +10.0 points higher for the 506 non-exact-discussion PRs. The 26 movement-only PRs sit highest at +38.8 points, on a very wide interval, which is one reason the active-discussion subset rather than this four-state model remains the primary specificity contrast. The gradient is also consistent with any visible activity by hour 48 marking a PR that is still being worked on, so it should not be read as an ordering of response types.

For the route model, automation with no user event is the reference. Controls include author product, reviewer product, source, month, trigger age, and the same pre-trigger activity measures. Standard errors cluster by repository. Table 14 gives the exact-edge sensitivities; Table 16 gives every route contrast; Table 17 gives the unmeasured-confounding bounds discussed in Section 9.3.

Neither model identifies an intervention. Task difficulty, maintainer intent, review meaning, and private activity can select both the public topology and later merge. The estimates are integration markers, not effects of adding an agent or user.

9.1 What the exposure and the cohort actually contain

Two properties of the RQ3 design are recoverable from the event tables and change how the estimate should be read, so Table 19 reports them explicitly.

First, the exposure rule requires a reply whose parent identifier is the trigger's own identifier. It does not require another product to write that reply. Of the 128 exposure events on the 109 exposed PRs, 105 are written by user accounts, 76 of them by the PR author's own account. Thirteen are written by the triggering product replying to its own comment, and three by a different mapped product. The exposure is therefore a public acknowledgement of an agent's review point, and in this cohort it is almost always a person who writes it.

Because of that composition, we refit the primary model under two stricter definitions on the same cohort. Removing replies written by the triggering product gives +15.9 percentage points (95% interval: 5.4 to 26.3) on 99 exposed PRs. Requiring a user-account reply gives +17.0 points (6.4 to 27.6) on 97. The estimate is not produced by a product answering itself. Only three PRs carry an exact reply from a different mapped product, which is too few to model; that count is the RQ1 scarcity result restated at the outcome stage.

Second, the landmark keeps PRs still open at hour 48. Of 3,942 cross-product inline-trigger PRs with a complete 30-day horizon, 2,875 close and 2,528 merge at or before hour 48. The 1,067-PR cohort is the 27% that remain. The estimates describe that slower-resolving group. For reply windows shorter than the landmark the exposure could in principle also influence whether a PR is still open at hour 48, which is a second reason to treat the 48-hour window as primary and the shorter windows as sensitivity analyses.

9.2 Two design-level extensions

The bounds in the next subsection all keep the hour-48 cohort fixed. Two further analyses change the design instead, and Table 20 reports both.

The first removes the cohort restriction. Every cross-product inline-trigger PR with a complete 30-day horizon is followed from its trigger. Follow-up is cut into eleven intervals, short at first and longer later, and each PR contributes one row per interval it survives. The exact addressed edge enters as a time-varying covariate: a PR is unexposed until its own first exact reply and exposed afterwards. This is what stops the model from crediting the edge with time that passed before the reply existed. The estimate is an odds ratio for merging in the next interval, from a pooled logistic model with interval indicators and repository-clustered standard errors. It stays near 1.7 across all three control sets.

The second asks who wrote the edge. Each exposed PR is labelled by the account that wrote its first exact reply, and that account is checked against the same strict prior-history rule used for RQ2: a submitted review by the same account, in the same repository, on a different PR, strictly before the trigger. The resulting split is large. It does not survive repository fixed effects. Prior reviewers and newcomers are mostly found in different repositories, and those repositories differ in how often anything merges. We therefore report the split as a description of where the association sits. It is not evidence that repository familiarity is what makes the reply work, and the article says so.

9.3 Bounding the unmeasured structure

Because the exact-edge contrast carries a headline number, we measure how much unmeasured structure would be needed to remove it instead of only warning that some could exist. Table 17 reports four checks on the same 1,067-PR cohort and the same pre-trigger specification.

Before those checks, Table 18 reports what the measured controls actually do. It gives the exposed mean, the unexposed mean, and the standardized mean difference for all six pre-trigger controls at all four reply windows. No absolute standardized difference exceeds 0.16. The largest values sit on log pre-trigger events, trigger age, and pre-trigger decisive reviews. The final block adds the propensity-score distribution of the 109 exact-edge PRs and the 506 non-exact-discussion PRs, so the reader can see the overlap directly. The two groups share most of their range, but the edge group extends further into high scores. Good balance on measured variables is a description of this sample. It is not evidence about task difficulty, maintainer intent, or anything else the public trace does not record.

First, the adjusted risk difference is rescaled to an approximate risk ratio using the unexposed later-merge rate, and the E-value is $\mathrm{RR} + \sqrt{\mathrm{RR}(\mathrm{RR}-1)}$. An unmeasured factor would have to be associated with both the exact edge and later merge at least this strongly, conditional on every measured control, before it could explain the result. The E-value exceeds 2.2 at all four reply windows.

Second, the tipping grid states the same requirement on the risk-difference scale without a ratio approximation. For one omitted binary pre-trigger factor, the induced bias is exactly the product of its prevalence difference between exposed and unexposed PRs and its own later-merge difference. The table reports the outcome difference required at two prevalence gaps.

Third, three negative-control outcomes are re-estimated with the same model. Each was already complete before the trigger, so the exposure cannot produce it and the expected estimate is null. Eleven of twelve models cover the null. The exception is the one-hour reply window against pre-trigger user activity (+16.7 points, 95% interval: 4.9 to 28.4), which is a genuine finding rather than noise: a reply inside one hour is selected by PRs that already had a person acting on them. That is why the primary window is 48 hours. At 48 hours the same placebo is +7.0 points (−1.7 to 15.7). We describe that as covering the null, not as a demonstrated zero: it is an underpowered residual of about seven points in the direction a sceptical reader would expect, and it is the main reason the measured controls should not be treated as sufficient.

Fourth, randomisation inference permutes the exposure label within each repository, holding that repository's exposed count fixed, and refits the model 2,000 times from a fixed seed. This gives a design-based reference distribution that does not rely on a clustered normal approximation with only 109 exposed PRs. Two versions are reported. The unconditional version in Table 17 permutes across the whole cohort, but its reference distribution is not centred at zero: 423 of 469 repositories contain no exposure variation, and without repository fixed effects the between-repository part of the coefficient is invariant to a within-repository permutation. The conditional version in Table 19 therefore restricts the fit to the 46 repositories that can actually be re-randomised and adds repository fixed effects. Its reference distribution is centred on zero, the observed within-repository estimate is +16.4 points, and the two-sided p is 0.007. The conditional version is the one quoted in the article; the unconditional version is retained so that the difference is on the record.

These are fragility bounds. A confounder above the reported strength would still overturn the association, and no check here converts it into a causal effect or into evidence that the review point was resolved. The measured controls also contain no proxy for task difficulty, so the E-value should be read as the strength required beyond the observed public activity record, not beyond everything that matters.

10 Structural review overlap and semantic coding gate

An exploratory cohort finds comments from two mapped reviewer products at the same original commit, path, and diff position. This is a structural overlap, not semantic duplication or complementarity. Table 21 reports the population, timing, concentration, and frozen gates.

The semantic gate is kept separate because designed multi-agent review already shows that structured disagreement and role separation can matter (Qiu and Gill 2026; Zhang 2026). Public co-location alone does not show the same mechanism.

Both blinded coder packets contain the complete structural population and omit product labels. Semantic categories remain empty. No product-general claim is allowed because one product pair exceeds the frozen concentration threshold. Even after coding, a headline requires agreement of at least $\kappa = 0.70$, no more than 30% unclear or boilerplate cases, and enough support for any named category. Automated timing and exact-text checks cannot replace this coding.

11 External attribution and construct checks

Table 22 records the evidence role assigned after schema, license, overlap, and construct checks. A large source does not enter the analysis merely because it contains agent activity. It must preserve the same event unit and observation horizon. Sources that preserve only text, counts, diffs, or private sessions remain construct checks rather than pooled replication data.

We joined the fixed landmark cohort to the independently released AI-to-AI cross-product cohort derived from CodAGE (Selvanayagam and Ghaleb 2026). The join uses repository, PR number, and exact product labels; it imports no outcome or post-trigger feature. Table 24 shows that product-pair attribution and trigger timing agree for nearly all observed overlap. This supports the event anchors, but not external generalization: the sources observe some of the same public GitHub PRs, and only nine overlapping PRs have an exact addressed edge. The wide outcome intervals are retained as an appendix sensitivity and not used as a headline.

We downloaded and scanned the full pinned SWE-Review-Chat release (Zhong et al. 2026). Table 23 applies the same exact product mapping, inline-parent rule, AIDev-overlap exclusion, and 48-hour landmark gate. Every pre-exclusion candidate PR already occurs in the full AIDev corpus, so no independent cohort remains. Before exclusion, nested children were user accounts or the same reviewer product; none was a different mapped product. The authenticated REST gate passed, but no request for comment or PR content was made because there was no disjoint candidate to hydrate. Zero disjoint support is a failed replication gate, not a claim that public connections cannot occur.

Two further artifacts were kept only for semantic measurement. SWE-PRBench contains real merged PRs and human-written initiating feedback (Kumar 2026), but it removes the usable reply path, uses synthetic per-PR comment identifiers, and supplies no event time or review-batch identifier. Its annotation files and PR-level summary counts also disagree, so the annotation files are the only accepted source of record. The AIReviewAction artifact linked to the developer-response study (Cynthia et al. 2026) has mixed comment and file-fragment grain. Its actionability and response labels are model-derived, with a small double-human audit, and its "addressed" label means an inferred later code change rather than an exact public reply. Its full-label lineage is also incomplete. A leave-one-source-out text-transfer check failed its frozen generalization gate, so no transferred label enters the AIDev analysis. These sources may inform a future codebook; they do not validate the three RQs.

12 Data quality, leakage, and robustness gates

Table 25 lists the load-bearing validation gates. Event order must be strictly after the trigger and inside its declared window. Each PR must have one state per threshold. Review batches must be distinct. Prior-history rows must exclude the focal PR and all future or equal timestamps. Landmark features must occur before hour 48, and outcomes only after it.

Nine exact duplicate force-push rows occur on two PRs. The analytical event view deduplicates them. The first-state assignment is identical with and without the surplus rows. Ordinary `committed` and `reviewed` timeline events have no usable timestamps in this release, so they are excluded from temporal claims. A force-push is visible branch movement, not a verified repair.

Uncertainty resamples repositories because PRs in one repository share policy, integrations, and contributors. Main ordering statements are also recomputed after removing each repository and each author–reviewer product pair. These checks test concentration; they do not remove all selection bias.

13 Variable dictionary

Table 26 lists every column of the derived landmark cohort with its type, its prose definition, and the leakage tag stored beside it. The tag has three values: available at or before the trigger, available after the trigger by hour 48, or an outcome read only after hour 48. This is the column-by-column form of the pre-trigger-only claim. The primary specification draws only on columns tagged at or before the trigger. Columns tagged after the trigger appear only in the secondary route decomposition and in the exposure itself. Columns tagged as outcome are never used as inputs. The table records where each value can be observed. It does not claim that any value is a good measure of the thing it is named after.

14 Experiment disposition

Table 29 records which attractive exploratory stories were kept, moved to this appendix, or rejected. Results are rejected when their construct cannot be observed, a stronger comparison removes the pattern, or a placebo reproduces it. This policy prevents the paper from selecting only positive estimates.

15 Reproduction contract

The reproducibility bundle pins the dataset revision; creates cross-product triggers, response events, and landmark cohorts; runs all robustness, falsification, and sensitivity analyses; generates the five main figures, the two appendix figures, and this appendix's tables; executes the automated test suite; compiles both PDFs; and scans the LaTeX logs. Public release must exclude private coder keys and any restricted raw material.

Table 28 gives that sequence step by step, with what each step produces and the frozen row count of one of its products where a stable count exists. The order is read from the repository README, so the table cannot drift from the documented commands. A rerun that changes a listed count is a signal to compare inputs rather than to accept the new number. The project requires Python 3.11 or newer, and `uv.lock` pins the fully resolved dependency graph, so `uv sync` installs the same package versions that produced these tables. The table records the order and the sizes. It does not certify that a rerun on a different dataset revision would give the same estimates.

Uncertainty statements in this paper come from two different machines, and the settings are not uniform, so Table 27 lists them per analysis: the resampling unit, the number of draws, the seed, and the interval method. Cluster bootstraps resample whole repositories, but the draw count is 10,000 for the matched contrasts and the same-locus timing share and 5,000 for the burst, deep-transition, and persistence shares. Order placebos permute event order inside a PR 500 times. Both randomisation-inference tests permute exposure labels inside repositories 2,000 times. Every seed derives from one fixed value, with fixed offsets so that separate quantities do not reuse one draw sequence. The outcome regressions use no resampling at all. Their intervals come from a clustered normal approximation, which is why the exact-edge estimate is checked by permutation as well.

External raw packages are also excluded from the release. Their revisions, licenses, sizes, and integrity checks are recorded in a frozen acquisition manifest. Separate profilers recreate the tracked aggregate audits when the pinned third-party downloads are available. The standard paper build consumes those aggregate records and does not make network calls or silently replace a missing external field with a negative event.

Every material claim should be read with its observation level. Public product presence is not private communication. Temporal succession is not semantic response. Account history is not verified memory retrieval. Merge is an integration state, not code quality. These boundaries are part of the result, not only limitations.

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